KEYNOTE-024: Clinical Deconstruction & Scientific Narrative

Pembrolizumab Monotherapy vs. Chemotherapy in First-Line PD-L1–High Advanced NSCLC

Alex Maria Younes, MBioMedSci (Medical Affairs Portfolio Exercise)

N Engl J Med 2016; 375:1823–1833 | ClinicalTrials.gov: NCT02142738

Thoracic Oncology Medical Communications | Clinical Study Review

Slide 1: Clinical Landscape & Frontline Unmet Need

First-Line Chemotherapy Offers Limited Long-Term Survival in Advanced NSCLC, Mandating Biomarker-Driven Regimens

The Clinical Status Quo Prior to Checkpoint Blockade:

- **Global Disease Burden:** Lung cancer remains the primary cause of cancer death worldwide; NSCLC accounts for ∼85% of cases, typically diagnosed at Stage IV.
- **Therapeutic Ceiling of Platinum Duos:**
  - Standard carboplatin/cisplatin doublets produce median PFS of **4–6 months**.
  - Median Overall Survival plateaus at **10–12 months**; 5-year OS $< 5\%$.
- **High-Volume Cytotoxic Morbidity:** Cumulative myelosuppression, nephrotoxicity, and intractable nausea limit treatment duration.
- **The Precision Therapy Gap:** Targeted TKIs exist for *EGFR/ALK* alterations (<15%); the remaining ∼**85% lack targetable driver oncogenes**.

HISTORICAL BENCHMARKS

4–6 Months

Historical Median PFS (Chemotherapy)

<5% 5-Year OS

Long-Term Survival Ceiling (Unselected NSCLC)

∼85% Wild-Type

Advanced NSCLC Patients Without Driver Mutations

Source: Reck M, et al. N Engl J Med 2016; 375:1823–33.

Slide 2: Biological Mechanism of Action (MOA)

Tumours Exploit the PD-1/PD-L1 Pathway for Immune Evasion; Pembrolizumab Restores Endogenous Cytolytic T-Cell Attack

A. TUMOUR IMMUNE EVASION

- Tumour cells upregulate surface **PD-L1** in response to endogenous interferon-γ signals.
- PD-L1 binds **PD-1 receptors** on tumour-infiltrating CD8+ cytotoxic T-lymphocytes.
- Transmits an inhibitory intracellular signal, triggering **T-cell exhaustion, anergy, and apoptosis**.

B. PEMBROLIZUMAB CHECKPOINT BLOCKADE

- High-affinity, **humanized IgG4 monoclonal antibody**.
- Selectively caps PD-1; blocks interaction with **PD-L1 and PD-L2**.
- Engineered IgG4 isotype **avoids ADCC** (prevents cytotoxic depletion of endogenous T-cells).
- Disinhibits anti-tumour immunity, restoring **T-cell proliferation and cytolytic granule release**.

The Biomarker Hypothesis: High tumour surface PD-L1 expression (TPS $\geq 50\%$) identifies tumours actively suppressing an endogenous T-cell immune response—creating an optimal primed substrate for checkpoint disinhibition.

Slide 3: KEYNOTE-024 Study Design & Biomarker Enrichment

Randomized Phase III Design Evaluated Pembrolizumab Monotherapy vs. Standard Chemotherapy in PD-L1 TPS $\geq 50\%$ NSCLC

Trial Flow & Treatment Architecture (N = 305):

- **Patient Eligibility:** Advanced/metastatic Stage IV NSCLC; Treatment-naïve; ECOG PS 0–1; No sensitizing *EGFR* mutations or *ALK* translocations.
- **Biomarker Gate:** PD-L1 Tumour Proportion Score (TPS) $\geq 50\%$ assessed by central laboratory via **IHC 22C3 pharmDx assay**.
- **1:1 Randomisation Architecture:**
  - **Pembrolizumab Arm (n = 154):** Fixed 200 mg IV q3w for up to 35 cycles (2 years) or until progression/intolerable toxicity.
  - **Chemotherapy Arm (n = 151):** Platinum doublet (carboplatin/cisplatin + pemetrexed/gemcitabine/paclitaxel) + optional pemetrexed maintenance.
- **Ethical Cross-Over Protocol:** Progression on chemo allowed switch to pembrolizumab (**43.7% on-study cross-over**).

BIOMARKER STRATEGY

Why TPS $\geq 50\%$ Threshold?
In early KEYNOTE-001 data, TPS $\geq 50\%$ marked the biological inflection point where ORR tripled ($\sim$45% vs $\sim$15% in TPS $<$50%). Enriched for highest-probability cohort to prove monotherapy superiority over doublet chemotherapy.

Clinical Prevalence:
1,945 screened $\rightarrow$ **30.2% (500/1,653)** had TPS $\geq 50\%$ (1 in 3 frontline patients).

Source: Reck M, et al. N Engl J Med 2016; 375:1823–1833.

Slide 4: Primary Endpoint: Progression-Free Survival (PFS)

Pembrolizumab Monotherapy Doubled Median PFS to 10.3 Months, Halving the Risk of Disease Progression or Death

BLINDED INDEPENDENT REVIEW (RECIST v1.1)

Pembrolizumab: 10.3 mo (95% CI: 6.7–NR)
Chemotherapy: 6.0 mo (95% CI: 4.2–6.2)

HR = 0.50 (95% CI: 0.37–0.68) **P < 0.001**

Landmark PFS Rates (6 & 12 Months):

- **6-Month PFS Rate:** 62.1% (pembro) vs. 50.3% (chemo)
- **12-Month PFS Rate: 48.0%** vs. **21.2%** (more than double)

KEY CLINICAL INSIGHTS

- **Early & Widening Separation:** Kaplan-Meier curves diverge at first restaging scan (Week 9) and separation expands through Month 18.
- **Subgroup Consistency:** Hazard ratio favoured pembrolizumab across all pre-specified baseline subgroups:
  - Age: $<$65 (HR 0.61) and $\geq$65 (HR 0.45)
  - Histology: Non-squamous (HR 0.55) and Squamous (HR 0.35)
  - Performance Status: ECOG 0 (HR 0.50) and ECOG 1 (HR 0.52)
- **Early IDMC Unblinding:** Due to overwhelming PFS superiority, early trial unblinding was recommended by the IDMC.

Slide 5: Secondary Efficacy: Overall Survival (OS)

Pembrolizumab Conferred a Significant 40% Reduction in Mortality Despite Substantial Control-Arm Cross-Over

INTERIM OVERALL SURVIVAL ANALYSIS

HR = 0.60 **(95% CI: 0.41–0.89)**

P = 0.005 (40% Mortality Reduction)

6-Month OS Rate: 80.2% (pembro) vs. **72.4%** (chemo)

12-Month OS Rate: 70.3% (pembro) vs. **54.0%** (chemo)

Median OS not reached in pembro arm at interim analysis; median follow-up 11.2 months.

CROSS-OVER DILUTION SENSITIVITY

The Intention-to-Treat Challenge:

- **43.7% (66/151)** chemotherapy patients crossed over to pembrolizumab on-study upon progression.
- Additional **7.9% (12 patients)** received off-study anti-PD-(L)1 therapy (>50% effective cross-over).
- In standard ITT analysis, control patients receiving salvage immunotherapy artificially improve the control arm curve, compressing the observed HR.

RPSFT Counterfactual Adjustment:

Adjusting for cross-over using Rank-Preserving Structural Failure Time modeling reveals an un-confounded HR of $\sim$**0.49**.

Clinical Takeaway: Frontline initiation is essential; delayed second-line use cannot fully recover lost survival benefit.

Slide 6: Secondary Efficacy: Response Rate & Duration

Pembrolizumab Produced Significantly Higher Objective Responses with Unprecedented Response Durability

OBJECTIVE RESPONSE RATE (ORR)

Pembrolizumab: 44.8% (95% CI: 36.8–53.0)
Chemotherapy: 27.8% (95% CI: 20.8–35.7)

Difference: +17.0% **(P = 0.0011)**

Depth of Response:
- Complete Response (CR): **3.9% (6 pts)** vs. 0.7% (1 pt)
- Partial Response (PR): **40.9% (63 pts)** vs. 27.2% (41 pts)
- Stable Disease (SD): 26.6% vs. 44.4%

DURATION OF RESPONSE (DOR)

Pembrolizumab: NOT REACHED (1.9+ to 14.5+ mo)
Chemotherapy: 6.3 Months (2.1+ to 12.6+ mo)

Kinetics & Persistence of Response:
- **Rapid Onset:** Median time to response was **2.2 months in both arms** (dispelling myth that immunotherapy is slow to act).
- **Immunological Memory:** At data cutoff, **88% of responders** on pembrolizumab remained in ongoing response vs. only **54%** on chemotherapy.

Slide 7: Safety Profile & Active imAE Management

Pembrolizumab Halved Severe High-Grade Toxicities, Featuring a Manageable and Reversible Immune Safety Profile

OVERALL TRAE SUMMARY

Adverse Event Metric	Pembro (n=154)	Chemo (n=150)
Any-Grade TRAEs	73.4%	90.0%
Grade 3–5 TRAEs	**26.6%**	**53.3%**
Discontinuations for TRAE	7.1%	10.7%
Treatment-Related Deaths	**0% (0 pts)**	**2.0% (3 pts)**

Elimination of Cytotoxic Toxicities:
Nausea (10.4% vs 43.3%) Anemia (5.2% vs 34.0%)
Neutropenia (0% vs 13.3%) Alopecia (0% vs ∼30%)

IMMUNE PNEUMONITIS MANAGEMENT

Incidence: Overall: **5.8% (9 pts)**; Grade 3–5: **2.6% (4 pts)**; 0 deaths.

Protocol Action Algorithm:

- **Grade 1 (Asymptomatic):** Monitor; repeat chest CT at 2–3 weeks.
- **Grade 2 (Dyspnea/Cough):** Withhold pembro; rule out infection; oral prednisone (1–2 mg/kg/day); **taper slowly over $\geq$4 weeks** to prevent rebound.
- **Grade 3–4 (Hypoxia/Severe):** Permanently discontinue; IV methylprednisolone (1–2 mg/kg/day); hospitalize.

Reassurance: Reversible in $>$80% without blunting efficacy.

Slide 8: Clinical Conclusions & Paradigm Shift

KEYNOTE-024 Redefined First-Line Care: Pembrolizumab Monotherapy Established as Standard in PD-L1 TPS $\geq 50\%$ NSCLC

EFFICACY REDEFINED

- **PFS Doubled:** 10.3 vs 6.0 mo (HR 0.50, $P < 0.001$)
- **OS Advantage:** HR 0.60 despite $>50\%$ cross-over
- **Durable Responses:** Median DOR not reached; 88% ongoing

TOLERABILITY ELEVATED

- **Halved Severe Events:** Grade 3–5 TRAEs 26.6% vs 53.3%
- **Chemo Morbidity Reduced:** Zero alopecia / neutropenic sepsis
- **Manageable imAEs:** Algorithmic steroid management

DIAGNOSTIC MANDATE

- **Upfront Reflex Testing:** Mandatory PD-L1 IHC 22C3 alongside EGFR/ALK
- **1 in 3 Patients:** TPS $\geq 50\%$ identifies frontline chemo-free candidates

Practice Paradigm: Chemotherapy-free first-line management with single-agent pembrolizumab is established as the preferred Category 1 NCCN / ESMO standard of care for PD-L1 TPS $\geq 50\%$ advanced NSCLC.